

Seminar

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1 Introduction: The Need for Certifiable Robustness

- Neural networks are vulnerable to small perturbations of the input.
- These perturbations are often imperceptible to humans but can completely change model predictions.
- Adversarial examples revealed fundamental weaknesses in deep neural networks.
- The discovery of adversarial examples motivated research on robust and verifiable machine learning.
- The distinction between empirical and certified robustness

References:

- Goodfellow et al. (2014) – *Explaining and Harnessing Adversarial Examples*
- Madry et al. (2017) – *Towards Deep Learning Models Resistant to Adversarial Attacks*

2 Foundational Concepts: Robustness Verification

- Formal definition of robustness: $\forall \delta \in S : f(x + \delta) = y$
- The verification problem as computing output bounds for input set S
- modern approaches use Co-Designs : training models in a way to encourage them to be simultaneously robust and easy-to-exactly-verify.
- for that verification algorithms get used while training

- Minimax robust optimization formulation: $\min_{\theta} \mathbb{E}[\max_{\delta \in S} L(f_{\theta}(x + \delta), y)]$
- Certified training minimizes a tractable upper bound on the inner maximization
- Trade-off: tighter bounds vs. computational efficiency vs. training stability

References:

- Wong and Kolter (2018) – *Provable Defenses via the Convex Outer Adversarial Polytope*
- Zhang et al. (2018) – *CROWN: A General Framework for Certifying Neural Network Robustness*
- Kai Y. Xiao et al. (2019) – *Training for faster adversarial Robustness Verification via Inducing ReLU Stability*
- Salman et al. (2020) – *Provably Robust Deep Learning via Adversarially Trained Smoothed Classifiers*
- Madry et al. (2017) – *Towards Deep Learning Models Resistant to Adversarial Attacks*
- Wong and Kolter (2018) – *Provable Defenses via the Convex Outer Adversarial Polytope*
- Gowal et al. (2018) – *On the Effectiveness of Interval Bound Propagation for Training Verifiably Robust Models*

3 Interval Bound Propagation (IBP)

- IBP propagates lower and upper bounds through each layer of a neural network.
- This allows verification of robustness within a perturbation interval.
- IBP is computationally efficient and scalable.
- However, the generated bounds may become loose in deeper networks.

Advantages:

- Fast
- Scalable
- Efficient training

Limitations:

- Lower bound precision / loose bounds
- Training instability

References:

- Gowal et al. (2018) – *On the Effectiveness of Interval Bound Propagation for Training Verifiably Robust Models*
- Mao et al. (2024) – *Understanding Certified Training with Interval Bound Propagation*

4 CROWN

- Linear relaxation methods approximate nonlinear activations with linear bounds.
- CROWN computes tighter robustness bounds compared to pure IBP.
- These methods generally achieve stronger certified robustness guarantees.

Advantages:

- Tighter bounds
- Improved certified accuracy

Limitations:

- Higher computational cost
- Increased memory consumption
- can slow training down about a factor of 100

References:

- Zhang et al. (2018) – *CROWN: A General Framework for Certifying Neural Network Robustness*
- Xu et al. (2020) – *Auto-LiRPA: A Unified Framework for Automatic Linear Relaxation Based Perturbation Analysis*

5 CROWN-IBP: Stable and Efficient Certified Training

- The CROWN-IBP method combines:
- efficient IBP bounds

- tighter CROWN bounds

Goal:

- scalable verified training
- stable optimization
- stronger certified robustness

Core idea:

- Use IBP for efficient bound propagation during training.
- Use CROWN to tighten final robustness bounds.
- Interval Bound Propagation (IBP) and CROWN were originally proposed as verification techniques for certifying neural network robustness.
- both methods were incorporated into certified training frameworks,
- computed robustness bounds are used to construct a differentiable robustness loss
- enables the optimization process to improve not only classification accuracy but also provable robustness against bounded input perturbations
- a parameter β is scheduled from 1 to 0, transitioning from tight CROWN-based bounds which stabilize the initially loose and unstable IBP bounds to highly efficient IBP-based bounds for final training (ramp up phase)

Additional techniques:

- epsilon scheduling
- loss balancing
- bound tightening

Expected benefits:

- faster training
- better scalability
- improved certified accuracy

References:

- Zhang et al. (2020) – *Towards Stable and Efficient Training of Verifiably Robust Neural Networks*
- Xu et al. (2020) – *Automatic Perturbation Analysis for Scalable Certified Robustness and Beyond*

6 Recent Developments

- Recent work focuses on improving scalability, tighter bounds, and faster certification.
- Modern verification methods extend the original CROWN and IBP frameworks.
- Branch-and-bound techniques improve bound precision for complete verification.
- Automatic linear relaxation frameworks simplify certified robustness analysis.

Examples:

- α, β -CROWN winner of recent VNN-COMP (International Verification of Neural Networks Competition) from 2021 to 2025
- auto.LiRPA
- Branch-and-Bound verification
- Fast certified training methods

References:

- Wang et al. (2021) – *Beta-CROWN: Efficient Bound Propagation with Per-neuron Split Constraints for Complete and Incomplete Neural Network Verification*
- Xu et al. (2020) – *Auto-LiRPA: A Unified Framework for Automatic Linear Relaxation Based Perturbation Analysis*
- Bunel et al. (2020) – *Branch and Bound for Piecewise Linear Neural Network Verification*
- Shi et al. (2022) – *Fast Certified Robust Training with Short Warmup*

7 Next Steps

- Read and summarize all major referenced papers.
- Compare IBP, CROWN, and adversarial training approaches.
- Analyze strengths and weaknesses of certified robustness methods.
- Investigate modern extensions of α, β -CROWN.
- look into real life application f.e. CertPHash
- Organize the final seminar report structure.

- Begin writing introduction and related work sections.

References

- Yang et al. (2025) – *CertPHash: Towards Certified Perceptual Hashing via Robust Training*